

A negative solution to MAIS-O87

Rob Sneiderman

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Abstract

For the finite-horizon coin-line model of MAIS-O87, we give a proof that every finite initialization of linear policy-gradient flow eventually orients the out-of-distribution probe correctly, at every fixed positive training diversity. The argument uses monotone parameter combinations, fixed eventual signs of adjacent-state logits, and exclusion of the all-positive adjacent pattern. It does not require convergence of training return to one.

Theorem 1. *Fix an integer $L \geq 4$ and a real $\varepsilon \in (0, 1]$. In the MAIS-A8 coin-line model, every finite initialization of linear policy-gradient flow satisfies*

$$a(t) + \lceil L/2 \rceil b(t) < 0 \quad \text{for all sufficiently large } t.$$

Consequently, for every $\sigma > 0$,

$$\lim_{t \rightarrow \infty} q_\varepsilon^{\text{RL}}(t; \sigma, L) = 0.$$

Thus the positive uniform lower bound asked for in O87 does not exist.

The theorem does not assert convergence of every policy probability or of the training return to one. Eventual correction of the probe is sufficient. The threshold time may depend on L , ε , and the initialization; no rate or uniform threshold is asserted.

1 Model and derivative notation

Write $w = (a, b, d)$, $H = 2L$,

$$z_{pc} = a + bp + dc, \quad r_{pc} = \frac{1}{1 + e^{-z_{pc}}}.$$

The position is $p \in \{0, \dots, L\}$; the training coin is $c \in \{1, \dots, L\}$, with probability

$$\mu_c = \frac{\varepsilon}{L} + (1 - \varepsilon)\mathbf{1}_{\{c=L\}} > 0.$$

Conditional on c , the initial position is uniform on $\{0, \dots, L\}$. An episode stops successfully on hitting c , including at time zero, and otherwise stops unsuccessfully after H steps. Left and right steps are clipped to $[0, L]$. These time-zero success and H -action conventions are those of [MAIS-A8](#), [Definition 3.1](#). The objective J is success probability, and $\dot{w} = \nabla J$ is Euclidean gradient ascent in these three coordinates.

Let J_c be conditional success probability. Temporarily regard all nonterminal logits for fixed c as independent variables, and define

$$y_{pc} = \text{sign}(c - p), \quad \kappa_{pc} = \left| \frac{\partial J_c}{\partial z_{pc}} \right|.$$

We use the identities

$$\frac{\partial J_c}{\partial z_{pc}} = y_{pc} \kappa_{pc}, \quad \kappa_{pc} > 0, \quad (1)$$

$$\nabla J = \sum_{c=1}^L \sum_{p \neq c} \mu_c \kappa_{pc} y_{pc} (1, p, c). \quad (2)$$

To prove (1), let $V_h(p, c)$ denote probability of hitting c within h steps, and let $\rho_t(p, c)$ be the probability, conditional on c , of occupying p at time t without having stopped. For fixed c , the Bellman recursion is

$$V_h(s, c) = (1 - r_{sc})V_{h-1}(s^-, c) + r_{sc}V_{h-1}(s^+, c) \quad (s \neq c),$$

with $V_0(s, c) = \mathbf{1}_{\{s=c\}}$ and $V_h(c, c) = 1$. Fix a nonterminal p and put $D_h(s) = \partial V_h(s, c) / \partial z_{pc}$. Differentiating one step yields

$$\begin{aligned} D_h(s) &= (1 - r_{sc})D_{h-1}(s^-) + r_{sc}D_{h-1}(s^+) \\ &\quad + \mathbf{1}_{\{s=p\}} r_{pc} (1 - r_{pc}) [V_{h-1}(p^+, c) - V_{h-1}(p^-, c)]. \end{aligned}$$

Here $D_0 = 0$ and $D_h(c) = 0$. Iterating this identity and averaging the initial state gives the occupancy weights ρ_t and hence

$$\frac{\partial J_c}{\partial z_{pc}} = r_{pc} (1 - r_{pc}) \sum_{t=0}^{H-1} \rho_t(p, c) [V_{H-t-1}(p^+, c) - V_{H-t-1}(p^-, c)], \quad (3)$$

where $p^+ = \min(p+1, L)$ and $p^- = \max(p-1, 0)$.

Below the coin, a path from p^- must hit p^+ before reaching c . The strong Markov property and monotonicity in the remaining horizon give $V_h(p^-, c) \leq V_h(p^+, c)$. Reflection gives the opposite inequality above the coin. At $h = H - 1$ the inequality is strict: the closer neighbor has positive success probability, whereas from the farther neighbor there is positive probability of avoiding the closer one for all h steps. This uses self-loops at a wall or back-and-forth steps away from the closer neighbor. Since $\rho_0(p, c) = 1/(L+1)$, equation (3) has the strict claimed sign. Equation (2) is the chain rule.

2 Escape, vanishing gradient, and monotone bounds

Set

$$A = d - b, \quad R = A - a, \quad S = A + a.$$

The flow is globally defined. Indeed, J is a finite polynomial in finitely many logistic probabilities of linear forms. Its gradient and Hessian are globally bounded, so the gradient field is globally Lipschitz and bounded. Equation (2) gives

$$A' = \sum_{c, p \neq c} \mu_c \kappa_{pc} |c - p| > 0, \quad \|w'\| \leq \sqrt{1 + 2L^2} A'. \quad (4)$$

If A had a finite upper bound, (4) would imply finite total arclength and convergence of w to a finite point. At that point A' is strictly positive, a contradiction. Therefore

$$A(t) \longrightarrow +\infty. \quad (5)$$

Also $J' = \|\nabla J\|^2$ and $J \leq 1$. Thus the squared gradient has finite integral. It is uniformly continuous, since the gradient and Hessian are bounded. A nonnegative, uniformly continuous, integrable function tends to zero; hence $\nabla J \rightarrow 0$. Equation (4) implies $A' \rightarrow 0$ and, because every μ_c is positive,

$$\kappa_{pc}(t) \rightarrow 0 \quad (p \neq c). \quad (6)$$

Finally,

$$R' = \sum_{c,p \neq c} \mu_c \kappa_{pc} (|c - p| - y_{pc}) \geq 0, \quad (7)$$

$$S' = \sum_{c,p \neq c} \mu_c \kappa_{pc} (|c - p| + y_{pc}) \geq 0. \quad (8)$$

Only the lower bounds $R \geq R(0)$ and $S \geq S(0)$, not divergence of R or S , are needed below.

3 Adjacent logits have fixed limiting signs

Define

$$\ell_c = z_{c-1,c} \quad (1 \leq c \leq L), \quad q_c = z_{c+1,c} \quad (1 \leq c < L).$$

Lemma 1. *Each adjacent logit tends either to $+\infty$ or to $-\infty$.*

Proof. At a left-adjacent state, retain $t = 0$ in (3). Right immediately succeeds. After choosing left, success requires a later right action at $c - 1$. Choosing left at every subsequent visit to $c - 1$ for $H - 1$ steps prevents success, with probability at least $(1 - r)^{H-1}$. Therefore

$$\kappa_{c-1,c} \geq \frac{r(1-r)^H}{L+1}. \quad (9)$$

Similarly, at a right-adjacent state,

$$\kappa_{c+1,c} \geq \frac{r^H(1-r)}{L+1}. \quad (10)$$

These estimates are independent of all other policy probabilities. Precisely, modify the policy at the adjacent state to choose deterministically away from the coin. Under the modified policy, hitting the coin from the away neighbor is impossible. The original probability of these paths is the expectation of r^N , or $(1-r)^N$, with $N \leq H - 1$. This also includes clipped wall actions.

On a compact interval of logits the right sides of (9)–(10) have positive lower bounds. Thus (6) makes the absolute value of each adjacent logit tend to infinity. Continuity then gives a fixed eventual sign. $\square$

4 The all-positive adjacent pattern is impossible

Lemma 2. *No trajectory has $\ell_c \rightarrow +\infty$ for every c and $q_c \rightarrow +\infty$ for every $c < L$.*

The lemma allows arbitrary signs at more distant training states.

4.1 Gradient estimates

Write $\nabla J_c = P_c - N_c$, where P_c is the contribution from $p < c$ and N_c is the magnitude of the contribution from $p > c$. Both are componentwise nonnegative by (1); $N_L = 0$. Let $F_c(p_0) = 1 - V_H(p_0, c)$. A path starting below the coin stops before visiting any state above it, so

$$P_c = \frac{1}{L+1} \sum_{p_0 < c} \nabla V_H(p_0, c) = -\frac{1}{L+1} \sum_{p_0 < c} \nabla F_c(p_0).$$

For the finite set $\mathcal{F}_c(p_0)$ of failed length- H action paths,

$$\nabla F_c(p_0) = \sum_{\omega \in \mathcal{F}_c(p_0)} \Pr_w(\omega \mid p_0, c) \nabla \log \Pr_w(\omega \mid p_0, c).$$

These identities make the failure-path score estimate below applicable to P_c . Put $n = \lceil (L+1)/2 \rceil \geq 3$. Constants $C, K > 0$ depending only on L satisfy the following estimates for every coordinate j . If $m_c = \min_{p < c} z_{pc} \geq 0$, then

$$(P_c)_j \leq C e^{-nm_c}. \quad (11)$$

For $c = 1$ and $h = z_{01} \geq 0$,

$$(P_1)_a = (P_1)_d \leq C e^{-Hh}, \quad (P_1)_b = 0. \quad (12)$$

If $q_c \geq 0$ and $c < L$, then, without restrictions on other logits,

$$(N_c)_j \geq K e^{-q_c}. \quad (13)$$

One may take $C = HL2^H$ and $K = 2^{-(H+1)}/(L+1)$.

For (11), a failed path starting below c stays below c . If it makes N_- left moves, its final position is at least $p_0 + H - 2N_-$; clipping at zero only increases this lower bound. Consequently $N_- \geq (H + p_0 - c + 1)/2$, and hence $N_- \geq n$. Each left probability is at most e^{-m_c} , and there are at most 2^H action sequences. The absolute score of a failed path in a parameter coordinate is at most HL . Differentiating and summing failure probabilities gives (11). Below-coin and above-coin dynamics decouple at the absorbing coin, so these are exactly the positive gradient contributions. For $c = 1$, failure from below requires H left self-loops at $p = 0$, proving (12). For (13), use (10), $r \geq 1/2$, and $r(1-r) \geq e^{-q_c}/4$. All components of $(1, c+1, c)$ are at least one.

4.2 Exclusion proof

Proof of Lemma 2. Suppose all adjacent logits tend to $+\infty$. At sufficiently late times $A > 0$ and $h = \ell_1 > 0$. Whenever $b \geq 0$, we have $d = A + b > 0$ and

$$m_c = h + (c-1)d, \quad q_1 = h + 2b.$$

Only $c \geq 2$ contribute positively to b' . With $\mu_1 = \varepsilon/L$,

$$b' \leq C e^{-n(h+d)} - \mu_1 K e^{-(h+2b)}. \quad (14)$$

The ratio of the positive term to the negative term is at most

$$\frac{C}{\mu_1 K} \exp(-(n-1)h - (n-2)b - nA),$$

which tends uniformly to zero as $A \rightarrow \infty$ for $h, b \geq 0$. Thus, at sufficiently late times, $b' < 0$ whenever $b \geq 0$. A first late crossing from $b < 0$ to $b \geq 0$ would occur at $b = 0$ with nonnegative left difference quotient, contradicting $b' < 0$ there. Hence no such crossing is possible.

If b remains nonnegative, it is bounded and decreasing. Now $d \rightarrow \infty$ and $h \rightarrow \infty$. The positive a and d components are bounded by

$$Ce^{-Hh} + Ce^{-n(h+d)},$$

while their negative contribution from $c = 1$ is at least $\mu_1 Ke^{-(h+2b)}$. Since b is bounded, both positive-to-negative ratios tend to zero. Thus $a', d' < 0$ eventually, contradicting $h = a + d \rightarrow \infty$.

Otherwise $b < 0$ eventually. For $c < L$,

$$m_c = \ell_c = q_c - 2b \geq q_c.$$

As $q_c \rightarrow \infty$, equations (11) and (13) give, in every coordinate,

$$(\nabla J_c)_j \leq -\frac{K}{2}e^{-q_c} \quad (15)$$

eventually. For $c = L$, use the exact identity

$$m_L = \ell_L = q_{L-1} + A.$$

Its positive gradient is at most $Ce^{-n(q_{L-1}+A)}$, which is eventually smaller than half the weighted negative contribution of coin $L - 1$ in (15). Every component of ∇J is therefore strictly negative eventually. This gives $q'_1 = a' + 2b' + d' < 0$, contradicting $q_1 \rightarrow \infty$. Both cases are impossible. $\square$

5 A late nonnegative probe forces positive adjacent logits

Let $k = \lceil L/2 \rceil \geq 2$ and $P = a + kb$. We show that along any sequence of times tending to infinity with $P \geq 0$, every adjacent logit tends to positive infinity. Write $R_0 = R(0)$ and $S_0 = S(0)$.

If $b \geq 0$, substituting $a = P - kb$ into $S = A + a \geq S_0$ gives

$$b \leq \frac{A + P - S_0}{k}.$$

Consequently

$$h = a + d = P + A - (k-1)b \geq \frac{A+P}{k} + \frac{k-1}{k}S_0 \geq \frac{A}{k} + \frac{k-1}{k}S_0. \quad (16)$$

For $A > 0$, $d = A + b > 0$, and every training logit satisfies $z_{pc} = h + bp + (c-1)d \geq h$. Equation (5) proves the claim in this case.

If $b < 0$, write $B = -b > 0$. Then $a = P + kB$, and $R = A - a \geq R_0$ gives

$$kB \leq A - P - R_0.$$

Writing $D = b + d = A - 2B$, we have

$$D = R + P + (k-2)B \geq R_0, \quad q_1 = P + A + (k-3)B.$$

For $k \geq 3$, $q_1 \geq A$. For $k = 2$,

$$q_1 \geq \frac{A}{2} + \frac{3P}{2} + \frac{R_0}{2} \geq \frac{A}{2} + \frac{R_0}{2}.$$

At sufficiently late times $A > 0$. In either case $q_1 \geq A/2 - |R_0|/2$. Since $q_c = q_1 + (c-1)D$,

$$q_c \geq \frac{A}{2} - L|R_0| \quad (1 \leq c < L). \quad (17)$$

The other adjacent logits satisfy $\ell_c = q_c + 2B$ for $c < L$ and $\ell_L = q_{L-1} + A$. They too diverge positively by (5). This completes the claim for both signs of b .

6 Completion and probability statement

Proof of Theorem 1. If $P(t) \geq 0$ at arbitrarily late times, the previous section supplies an unbounded sequence of times along which all adjacent logits are positive. Lemma 1 says that each has a fixed eventual sign and diverges in magnitude. All must therefore tend to $+\infty$, contradicting Lemma 2.

Thus $P(t) < 0$ for all sufficiently large t , from every finite initial vector. The indicator $\mathbf{1}_{\{P(t)>0\}}$ tends pointwise to zero and is bounded by one. Bounded convergence under $N(0, \sigma^2 I_3)$ gives $q_\varepsilon^{\text{RL}}(t; \sigma, L) \rightarrow 0$. The same conclusion holds for any probability law on finite initial vectors. $\square$

No interchange of diversity and time limits is used. At every fixed positive ε , the limsup in O87 is zero, directly ruling out a positive δ uniform over sufficiently small positive diversity.

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References

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